1

$A = \begin{pmatrix} 1 & 3 \\ 5 & 2 \end{pmatrix}, B = \begin{pmatrix} 4 & 9 \\ 2 & 6 \end{pmatrix}$

a. $A + B = \begin{pmatrix} 1+4 & 3+9 \\ 5+2 & 2+6 \end{pmatrix} = \begin{pmatrix} 5 & 12 \\ 7 & 8 \end{pmatrix}$

b. $A - B = \begin{pmatrix} 1-4 & 3-9 \\ 5-2 & 2-6 \end{pmatrix} = \begin{pmatrix} -3 & -6 \\ 3 & -4 \end{pmatrix}$

c. $AB = \begin{pmatrix} 1(4)+3(2) & 1(9)+3(6) \\ 5(4)+2(2) & 5(9)+2(6) \end{pmatrix} = \begin{pmatrix} 10 & 27 \\ 24 & 57 \end{pmatrix}$

d. $BA = \begin{pmatrix} 4(1)+9(5) & 4(3)+9(2) \\ 2(1)+6(5) & 2(3)+6(2) \end{pmatrix} = \begin{pmatrix} 49 & 30 \\ 32 & 18 \end{pmatrix}$

2

$A = \begin{pmatrix} 9 & 1 \\ 1 & 4 \end{pmatrix}, B = \begin{pmatrix} 2 & 6 & 5 \\ 3 & 1 & 8 \end{pmatrix}$

a. Non-conformable

b. Non-conformable

c. $AB = \begin{pmatrix} 9(2)+1(3) & 9(6)+1(1) & 9(5)+1(8) \\ 1(2)+4(3) & 1(6)+4(1) & 1(5)+4(8) \end{pmatrix} = \begin{pmatrix} 21 & 55 & 53 \\ 14 & 10 & 37 \end{pmatrix}$

d. Non-conformable

3

Show that the following is true for any invertible, square matrics A,B,C,D of the same size: $[(A+B)^\top D^{-1}C^{-1}]^{-1} = CD(A^\top + B^\top)^{-1}$

By the distribution of inverses (applying it twice), then distributing the transpose, we get: $[(A+B)^\top D^{-1}C^{-1}]^{-1} = (D^{-1}C^{-1})^{-1}[(A+B)^\top]^{-1} = CD[(A+B)^\top]^{-1} = CD(A^\top + B^\top)^{-1}$

4

$A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}, B = \begin{pmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{pmatrix}$

a. $AB = \begin{pmatrix} A_{11}B_{11} + A_{12}B_{21} & A_{11}B_{12} + A_{12}B_{22} \\ A_{21}B_{11} + A_{22}B_{21} & A_{21}B_{12} + A_{22}B_{22} \end{pmatrix}$

b. To tranpose a partioned matrix, you transpose each partion in the overall matrix, then each partion gets transposed in itself

$A^\top = \begin{pmatrix} A_{11}^\top & A_{21}^\top \\ A_{12}^\top & A_{22}^\top \end{pmatrix}$

5

$x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, A = \begin{pmatrix} a_{11} & a_{12} \\ a_{12} & a_{22} \end{pmatrix}$

Find $\frac{dq(x)}{dx}$ where $q(x) = x^\top Ax$

Treating this more like normal algebra, $x^\top Ax$ would be the equivalent to Ax^2, and $\frac{d}{dx}Ax^2 = 2Ax$

Thus, because A is symmetric: $\frac{dq(x)}{dx} = 2Ax = \begin{pmatrix} a_{11}x_1 + a_{12}x_2 \\ a_{12}x_1 + a_{22}x_2 \end{pmatrix}$